

Polymer Composites

- **Isotropic** – most **metals, plastics and ceramic** components are broadly isotropic.
- **Anisotropy** – with **composites**, processing route and final component properties are intertwined
- Properties depend on **material orientation**, hence the selected **processing route**.

Structure

- 3 constituent parts: **Fibres**, **Matrix**, **Interface**

→ **Fibres** include: glass, aramid (Kevlar), Carbon

→ **Matrix** include: thermoset + thermoplastic resins – PE, epoxy, PEEK, Polyamide

→ **Interface** altered by surface condition of fibres and processing conditions used for manufacture

Thermosetting Matrix:

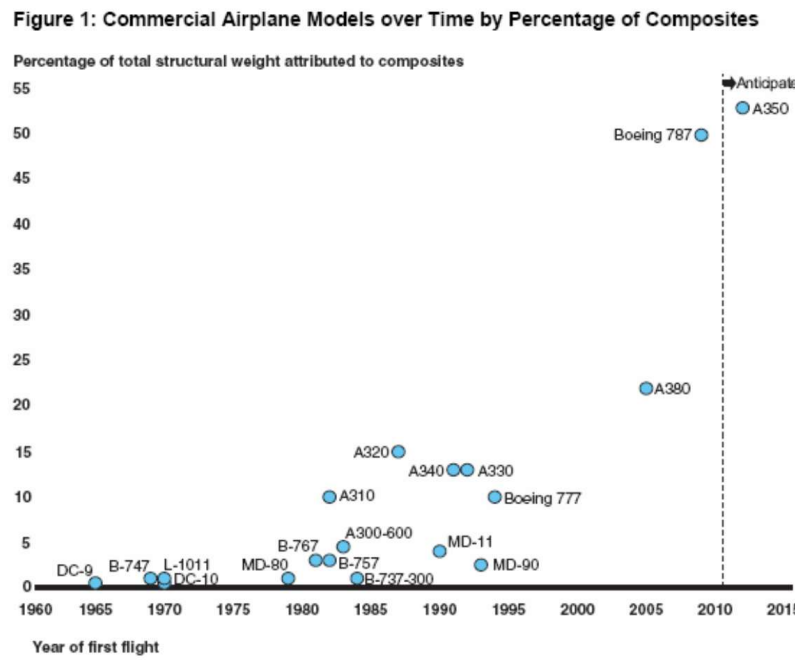
- Resin for matrix = **Low viscosity liquid**
- Resin **flows and combines** with solid fibres relatively **easily**.
- Fibres available in **different forms**
- Then **Matrix** undergoes **curing reaction**.

Common resin types:

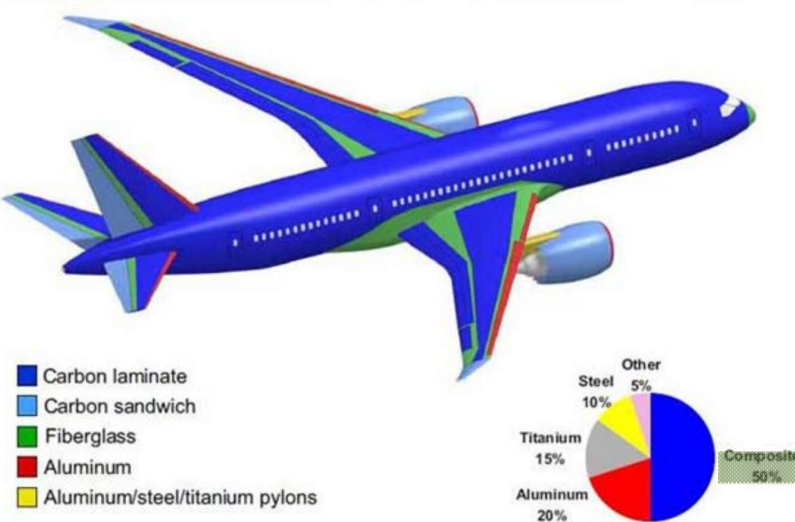
- Unsaturated polyesters
- Vinyl esters
- Epoxies
- Bismaleimides
- Polyimides phenolics
- Furanes
- Urethane – acrylates

Thermoplastic Matrix:

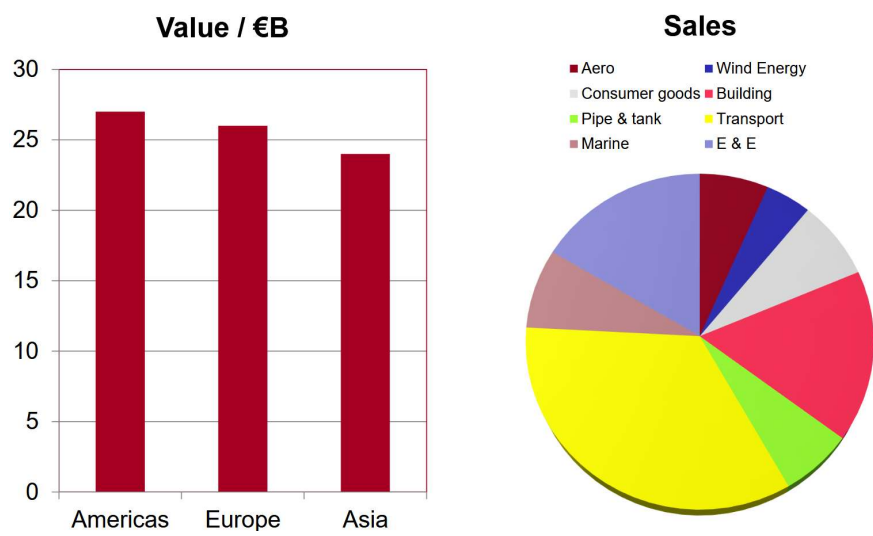
- Require **heat to soften** polymer matrix during processing
- The **melt viscosity** is typically **higher than the resin viscosity** of thermosets
- ∴ **Difficulty** in **impregnating** fibres with polymer; **historically used** in conjunction with **short chopped fibres**
- Resulting properties of composites are **limited**



Modern Aircraft Boeing 787 - Dreamliner



The Composites Industry



THE USA Composites market

- **American** composites market is **36% of the global composites industry** in value (22 billion Euros) and **35% in volume** (2.7 million metric tons).
- **Composites in the USA** represents **16% of the US structural materials market** (compared with **steel: 76%, aluminium: 8%**)
- The **volume of composites per capita** is **8-9kg**, compared to **1-2kg per capita in China**, for instance.

Continuous roving

- Relatively cheap + consist of 1000's of fibres which can be aligned.

Chopped glass fibre mats

- Low cost random fibre composites

Continuous filament fibre mats:

- Similar to chopped; slightly better properties.

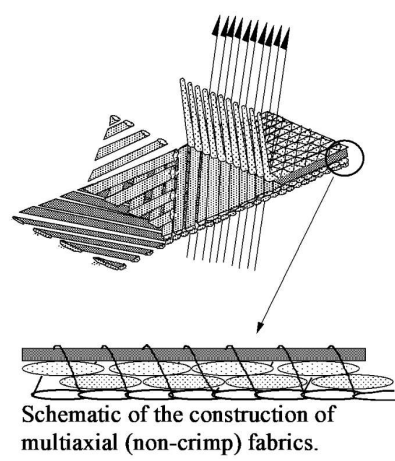
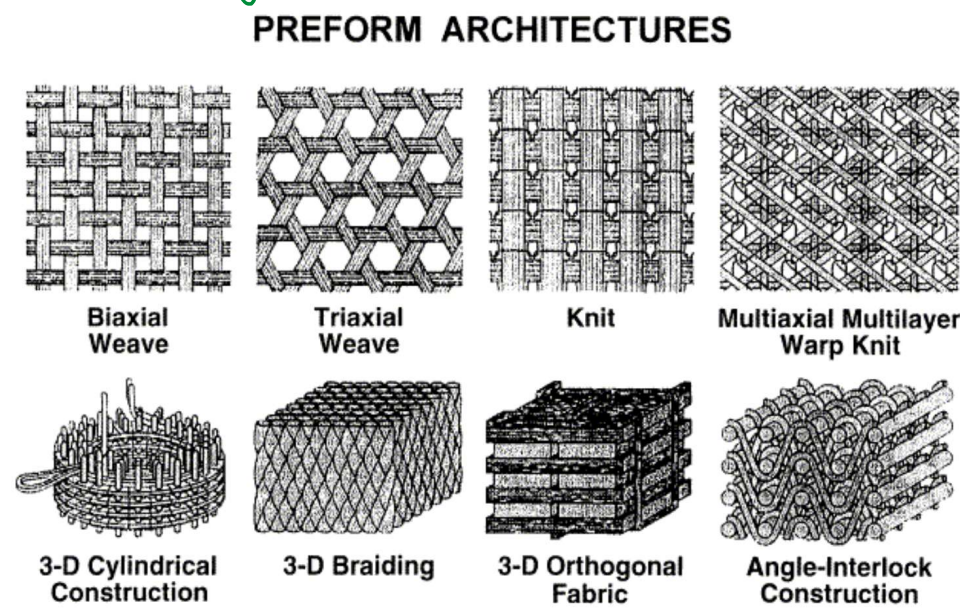
Woven fabrics:

- Higher Volume fraction, better properties

→ Triaxial Weaves

→ Adopting **non-woven technology** and **stitching** allows **unidirectional or multidirectional continuous fibre fabrics** to be produced **without fibre crimping**

→ **Knitting + braiding** techniques can produce **3D shaped preforms**



Contact Moulding – Hand Lay Up Methods

Hand Lay Up

Advantages of Hand lay-up:

- Simple
- Low capital costs
- Large size

Disadvantages of Hand Lay up:

- High labour content
- Low speed
- Quality dependant on operator skill when laying out fibres
- Only one 'good' surface
- Fibres may be damaged
- Difficult to accurately position fibres

Typical products types include:

- storage tanks,
- chemical plant,
- boat hulls, and
- train cabs

Roll Add resin Reinforcement
Hand lay-up Mould

Contact Moulding – Spray Up Methods

Spray Up

• Like hand lay up except the fibres and resin are deposited onto the mould by a spray system.

• Fibres are used in the form of continuous rovings, and the resin is mixed in the spray head.

Advantages of Spray up:

- Portable
- Uses rovings (cheaper than mats)
- Capital investment low
- Fast
- Reduced labour costs
- Accurate metering of catalyst accelerator

Disadvantages of spray up

- Only one 'good' surface
- Only useful for relatively large areas
- Only random fibres, and
- Uniformity very dependant on operator.

Typical products types include:

- Baths
- Cladding panels
- Large boat hulls, and
- Fish ponds.

Fibre chopper
Heads
Spray up Mould

Resin Transfer Moulding (RTM)

Advantages / Disadvantages of RTM

Disadvantages of RTM

- Worst strength and stiffness properties to those achieved using hand lay-up
- This results from low volume fractions of fibres in the final component and the poor wet out of the fibres.

Advantages of RTM

- Potential for high volume production
- Applicable to a wide range of sizes with low cost tooling
- Ability to combine different reinforcement forms easily with a degree of control on fibre orientation
- Ability to incorporate inserts
- Two good surfaces are produced

Resin Requirements

- Low viscosity during winding
- The pot life should be long
- The toxicity low

MAT602 – Materials Selection in Design

Filament Winding

Filament Winding

• The filament winding process produces high performance composites where the fibres are placed exactly where they will be required during the service life of the component.

The process limitations are mainly:

- a restriction on production rate,
- the formation of only one high quality surface, and
- the inability to wind into concave surfaces.

• The need to remove mandrels after winding certain shapes such as spheres is largely overcome by the use of inflatable, sacrificial or washout tooling.

• Uses: tubes, fishing rods, wind turbine blades, pole vaults.

Mandrel
Fibres of glass carbon or Kevlar
Resin bath
Winding
Mould containing form profile

Prepreg Autoclave Processing

Prepreg Autoclave Processing

• The prepregs are arranged on the tool by either a **hand lay-up** or a **tape lay-up machine** (similar in concept to filament winding).

• The prepregs typically retain their position on a tool during the lay-up process by virtue of their inherent "lack".

• The assembly is then completely consolidated and cured by placing it in an autoclave.

Hand lay-up machine
Tape lay-up machine
Autoclave

Pultrusion

Pultrusion

• Pultrusion is **analogous to extrusion** as used for metals and plastics.

• Continuous profile sections are produced by pulling fibres impregnated with resin through a heated steel die.

• This process involves an impregnation stage and a consolidation / forming operation.

• The rate of pultrusion is determined by the rate of curing of the resin within the heated tool.

• The section is then wound onto a drum or cut into required lengths.

• **USES:** beams, channels, ladders, bridges, corrosion resistant walkways.

Resin bath
Heated die
Pultrusion machine

Hot Press Moulding

Hot Press Moulding

• High volume production (>10,000 articles from one mould) can require the use of hot press (compression or injection) moulding in combination with specially formulated moulding compounds.

• For **mass production** the cycle time must be short at around a few minutes per article.

• The cure and the forming technique require high temperatures, which accelerates cure and reduces resin viscosity and the forming pressure.

• These materials must contain all of the ingredients for the final composite.

• The resin must be pre-accelerated and catalysed. The compound must also possess a reasonable storage life.

• **Fibres are also added to the compound to aid processing**

- by making the material **stiffer**
- reducing the shrinkage of the material after moulding and
- to reduce the cost

Moulding

Compression moulding (typically uses sheet moulding compound)

- Flash is difficult to remove (contains fibres).
- Charge difficult to meter and position reproducibly.

Injection Moulding (typically uses dough moulding compound):

- The mould is closed and compound introduced via runners.
- Mould/bolt clamping devices very expensive.
- Part size limited by machine considerations.
- Fibres suffer attrition during transport along screw of injection moulder and during passage through runners and die.
- Probability of alignment of fibres due to flow geometry may result in unfavourably orientated weak sections.

High Performance Composite Structures

High Performance Composite Structures

• High performance composites require continuous fibres positioned accurately in certain specified directions.

• The processes available depend on the initial form of the materials.

• For large and complex shapes the fibres and matrix usually come in a pre-combined sheet form known as prepreg which are processed in an autoclave or equivalent device.

